

AI Snake Oil: What Artificial Intelligence Can Do, What It Can't, and How to Tell the Difference

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1. Introduction to AI and Its Scope

Artificial intelligence (AI) is an umbrella term encompassing a broad spectrum of technologies that often share little in common apart from the label "AI." For instance, ChatGPT, which generates conversational text, operates fundamentally differently from the predictive software banks use to evaluate loan applicants. The diversity of AI applications creates a challenge for distinguishing between what AI can genuinely achieve and the exaggerated claims often made about it. This distinction becomes critical as AI continues to permeate various industries, requiring informed skepticism about its capabilities and limitations. Predictive AI, which forecasts future trends or behaviors, contrasts sharply with generative AI, which produces content such as text or images. Despite AI's promise, many applications remain unreliable, revealing inherent flaws and limitations in specific contexts. Moreover, the term "AI" itself is often misapplied due to historical usage and marketing influences, leading to misunderstandings about what qualifies as AI. These misrepresentations contribute to the phenomenon of "AI snake oil"—technologies that fail to live up to their claims. The book emphasizes the need for clarity and evidence to differentiate between genuinely functional AI and overhyped solutions.

2. Generative AI and Its Pitfalls

Generative AI, such as chatbots, has demonstrated significant utility in areas like research assistance, providing users with tools to explore unfamiliar domains. However, the effectiveness of these tools is often marred by their unreliability and the effort required to mitigate their shortcomings. ChatGPT exemplifies this dual nature by showcasing impressive capabilities while occasionally producing misleading or inaccurate outputs. The authors argue that the adaptation to generative AI demands not only technical skills but also critical thinking to navigate its pitfalls. AI-generated content can also inadvertently perpetuate misinformation or biases, highlighting the importance of scrutinizing its outputs. Additionally, the immense computational demands of models like ChatGPT illustrate the technical challenges inherent in scaling generative AI. Ethical concerns also arise, particularly regarding the working conditions of annotators tasked with training these models, who often face low pay and exposure to harmful content. Despite these challenges, generative AI continues to grow in prominence, necessitating robust frameworks for ethical use and evaluation.

3. Predictive AI and Its Limitations

Predictive AI, designed to forecast events or behaviors, faces its own set of challenges and limitations. The authors highlight numerous failed attempts at prediction, such as Google Flu Trends, which initially showed promise but ultimately failed due to data inconsistencies and unforeseen variables. Predictive AI often suffers from overfitting, where models perform well on training data but fail to generalize to new datasets. In social contexts, predictive AI frequently grapples with the complexity and unpredictability of human behavior, making its application in areas like hiring or healthcare fraught with ethical and technical challenges. Furthermore, many predictive AI claims lack rigorous validation, with companies often emphasizing non-discriminatory outcomes while neglecting performance accuracy. This tendency underscores the need for transparent evaluation methods and a cautious approach to adopting predictive AI in high-stakes scenarios.

4. Ethical and Societal Impacts

The book underscores the moral implications of deploying AI in areas such as hiring, healthcare, and criminal justice. For example, predictive models in healthcare have shown racial biases, as seen in an AI tool that underestimated healthcare needs for Black patients due to flawed assumptions about cost-based risk assessment. Similarly, AI-driven hiring tools can perpetuate inequalities or incentivize applicants to game the system with tactics like keyword

stuffing. The societal ramifications of flawed AI extend to amplifying systemic disparities and eroding trust in decision-making processes. The authors advocate for incorporating ethical considerations and fairness metrics into AI development and deployment, ensuring that AI technologies benefit all users equitably.

5. AI and Data Challenges

One of the fundamental issues with AI lies in the quality and representativeness of the data used to train models. The book argues that reliance on historical data can perpetuate existing biases, while the lack of context-specific datasets limits the applicability of AI in diverse settings. Collecting new, high-quality data is often expensive and time-consuming, discouraging companies from pursuing rigorous evaluation methods. Moreover, data leakage—when training and evaluation datasets overlap—can lead to inflated performance metrics, misleading users about AI's true capabilities. The authors call for greater transparency in data practices and emphasize the importance of randomized controlled trials (RCTs) as a benchmark for evaluating AI systems.

6. Cultural and Contextual Considerations

The application of AI across different cultural and geographical contexts presents unique challenges. Policies and algorithms often reflect the biases and values of their developers, leading to misalignments when deployed globally. Social media platforms, for example, enforce content moderation policies that may seem overly permissive or restrictive depending on local cultural norms. The authors highlight the need for context-aware AI systems that can adapt to diverse environments while maintaining ethical standards. However, achieving this balance requires significant investments in localized expertise and resources, which many companies are unwilling to make.

7. AI Regulation and Governance

The regulation of AI varies significantly across regions, with the United States favoring sector-specific rules, the European Union adopting horizontal regulations, and China emphasizing control aligned with state ideologies. These approaches reflect differing priorities, from privacy protection to ensuring national security. However, regulatory frameworks often struggle to keep pace with rapid technological advancements, creating gaps in oversight. The authors argue for a balanced approach that combines robust regulations with the flexibility to accommodate innovation. Strengthening regulatory agencies and fostering international cooperation are essential steps in ensuring that AI development aligns with public interest.

8. AI in Scientific Research

AI's integration into scientific research has been both transformative and problematic. While AI tools can accelerate data analysis and pattern recognition, they also introduce risks of reproducibility issues and methodological flaws. The authors cite numerous examples where AI-based studies failed to replicate, often due to data leakage or insufficient transparency. These shortcomings undermine trust in AI-driven research and highlight the need for rigorous validation protocols. The book calls for higher standards in publishing and peer review to ensure that AI research contributes reliably to scientific knowledge.

9. The Broader Impact of AI

Beyond specific applications, AI's influence on society raises profound questions about power dynamics and economic inequality. The authors contend that AI often centralizes power in the hands of a few companies, exacerbating existing disparities. For instance, AI tools designed to optimize efficiency can inadvertently marginalize vulnerable populations or reduce job opportunities. Addressing these challenges requires rethinking the incentives that drive AI adoption and exploring alternative models of governance and decision-making. The book suggests strategies like partial lotteries to introduce fairness and reduce systemic biases in resource allocation.

10. Looking Ahead

The authors conclude with a cautionary note about the future of AI, emphasizing the need to balance optimism with vigilance. While advancements in AI hold immense potential, they also bring risks that demand proactive management. The book calls for a multidisciplinary approach to AI development, incorporating insights from ethics, sociology, and policy-making to address its complex implications. By fostering transparency, accountability, and inclusivity, society can harness AI's benefits while mitigating its harms. Ultimately, the

authors advocate for a collective effort to navigate the challenges of AI, ensuring that it serves humanity's best interests.

Excerpts

Location: 173

Artificial intelligence, AI for short, is an umbrella term for a set of loosely related technologies. ChatGPT has little in common with, say, software that banks use to evaluate loan applicants. Both are referred to as AI, but in all the ways that matter—how they work, what they're used for and by whom, and how they fail—they couldn't be more different.

Location: 182

In contrast to generative AI is predictive AI, which makes predictions about the future in order to guide decision-making in the present.

Location: 189

Since AI refers to a vast array of technologies and applications, most people cannot yet fluently distinguish which types of AI are actually capable of functioning as promised and which types are simply snake oil.

Location: 243

We think most knowledge industries can benefit from chatbots in some way. We use them ourselves for research assistance, for tasks ranging from mundane ones such as formatting citations correctly, to things we wouldn't otherwise be able to do such as understanding a jargon-filled paper in a research area we aren't familiar with. The catch is that it takes effort and practice to use chatbots while avoiding their ever-present pitfalls.

Location: 279

But in the short term, adjusting to a world with generative AI is proving to be painful, as these tools are highly capable but unreliable.

Location: 338

We should be wary of predictive AI companies' claims unless they are accompanied by strong evidence.

Location: 359

In the end, whether an application gets labeled AI is heavily influenced by historical usage, marketing, and other factors. We won't fret about the fact that there's no consistent definition.

Location: 363

There's a humorous AI definition that's worth mentioning, because it reveals an important point: "AI is whatever hasn't been done yet." In other words, once an application starts working reliably, it fades into the background and people take it for granted, so it's no longer thought of as AI.

Location: 377

Accurately predicting people's social behavior is not a solvable technology problem, and determining people's life chances on the basis of inherently faulty predictions will always be morally problematic.

Location: 393

Facial recognition AI, if used correctly, tends to be accurate because there is little uncertainty or ambiguity in the task. Such AI is trained using vast databases of photos and labels that tell it whether or not any two photos represent the same person. So, given enough data and computational resources, it will learn the patterns that distinguish one face from another. Facial recognition is different from other facial analysis tasks such as gender identification or emotion recognition, which are far more error prone.

Location: 404

The biggest danger of facial recognition arises from the fact that it works really well, so it can cause great harm in the hands of the wrong people. Kashmir Hill, in her book *Your Face Belongs to Us*, details many harmful ways in which it has been used.²⁸ For example, oppressive governments can and do use it to identify people in peaceful protests and retaliate against them.

Location: 458

In the course on limits to prediction, students in the class were interested in predictive AI: in any and all attempts to predict the future using data, especially in social settings, ranging from civilizations to social media. Some interesting questions we looked at were: Can we predict geopolitical events such as election outcomes, recessions, or social movements? Can we predict which videos will go viral? What we found was a graveyard of ambitious attempts to predict the future.

Location: 464

The course included many case studies, including Google Flu Trends. This was a project that Google launched in 2008 to predict flu outbreaks by analyzing the search queries that its millions of users make every day. An increase in searches for flu-related terms could be indicative of an imminent outbreak. Google heavily promoted it as an example of AI and mass data collection used for social good. But within a few years, the accuracy of the predictions dropped precipitously. One reason was that it is hard to distinguish between media-driven panic searches and actual increases in flu activity. Another was that Google's own changes to its app changed people's search patterns in ways that weren't accounted for by the AI. Google Flu Trends ultimately ended up as a cautionary tale.

Location: 483

A 2023 paper claimed that machine learning could predict hit songs with 97 percent accuracy.

Location: 489

Unfortunately for music producers, we found that the study's results were bogus. The method presented in the paper exhibits one of the most common pitfalls in machine learning: data leakage. This means roughly that the tool is evaluated on the same, or similar, data that it is trained on, which leads to exaggerated estimates of accuracy.

Location: 493

We redid the analysis after fixing the error and found that machine learning performed no better than random guessing.

Location: 496

Systematic reviews of published research in many areas have found that the majority of machine-learning-based research that was re-examined turned out to be flawed.

Location: 507

We ourselves discovered flaws in many studies, mainly in the field of trying to predict civil wars (in short: it doesn't work).

Location: 513

The party line among scientists is that science "self-corrects," meaning that the normal process of science is sufficient to root out flawed research, but everything we've seen about the process suggests otherwise.

Location: 531

Notably, as far as we know, no hiring automation company has ever published a peer-reviewed paper validating its predictive AI, or even allowed an external researcher to evaluate it. Two of the leading companies made a show of external audits: Pymetrics contracted with a leading research group from Northeastern University, and HireVue contracted a noted independent auditor. But in both cases, the researchers were allowed to analyze only whether the AI was biased with respect to race or gender, and not whether it worked. The companies cleverly used a concern about discrimination to their advantage. If what you have is an elaborate random number generator that works equally poorly for everyone, it's easy to make it unbiased!

Location: 583

AI snake oil is AI that does not and cannot work, like the hiring video analysis software that originally motivated the research that led to this book. The goal of this book is to identify AI snake oil—and to distinguish it from AI that can work well if used in the right ways.

Location: 598

FIGURE 1.2. The landscape of AI snake oil, hype, and harms, showing a few illustrative applications.

Location: 672

More broadly, we show how concerns about AI, especially in the labor market, are often really about capitalism. We must urgently figure out how to strengthen existing safety nets and develop new ones so that we can better absorb the shocks caused by rapid technological progress and reap its benefits.

Location: 789

an assumption built into much of predictive AI: people with similar characteristics will behave similarly in the future.

Location: 841

Generally, collecting data is expensive and time-consuming. So, some AI developers claim that existing data is enough to make good decisions. The importance of collecting new data is widely recognized in medicine, which relies on randomized controlled trials (RCTs) to test the impact of introducing a new drug or vaccine. To test if a drug performs well, half the people in the trial are randomly assigned to the experimental group and are given the drug. The other half are assigned to the “control” group and are given a placebo. The groups are carefully balanced with respect to demographics like age and gender. A drug’s effectiveness is evaluated based on whether it leads to lower disease rates in the experimental group compared to the control. Medical researchers perform RCTs despite their slow pace and high expense for a simple reason—easier, faster methods don’t work. The same is true in many areas where AI is used for automated decision-making. Unfortunately, predictive AI companies don’t seem to have realized the importance of collecting data on the impact of their tools’ decisions. And such data collection is also expensive and time-consuming, so it would negate their claims of cutting costs and increasing efficiency. As a result, even if AI can make accurate predictions based on past data, we can’t know how good the resulting decisions will be before AI is deployed on a new dataset or in a new setting.

Location: 869

candidates have developed strategies to work around opaque hiring AI. They stuff their résumés with keywords from the job application and add the names of top universities in white text (which a human reader can’t see, but a computer can recognize).²² In video interviews that they know will be judged by AI, they try using fancy words such as “conglomerate.”

Location: 945

a predictive AI system that performs well in one country could be utterly useless in another.

Location: 975

The cost of flawed AI is not borne equally by all. The use of predictive AI disproportionately harms groups that have been systematically excluded and disadvantaged in the past.

Location: 996

The tool had a racial bias because the predicted quantity (healthcare costs) was a poor stand-in for what the developers claimed to predict (healthcare needs or a patient’s risk level).

Location: 1,047

the discomfort we experience with randomness can lead to a search for patterns where none exist. This paves the way for biases.

Location: 1,064

Reason Example A good prediction can result in a bad decision. Patients with asthma could be sent back home when they come to a hospital with symptoms of pneumonia. People can strategically game opaque AI. Adding bookshelves in the background increases scores on automated hiring tools. Users over-rely on AI without adequate oversight or recourse. The Dutch welfare fraud detection model falsely accused 30,000 parents of fraud without any recourse. Data for training AI may come from a different population than the one it is used on. PSA's criminal risk prediction relied on a national sample. It overestimated the risk in counties where crime was rarer. Predictive AI can increase inequality. Optum's Impact Pro led to an increase in the difference in the quality of care between Black and White patients.

Location: 1,175

Broadly speaking, machine learning is more suited for predicting things about individuals whereas simulation is more suited for predicting collective or global outcomes. That's because there are millions of people about whom you can collect training data to build machine learning models. For example, spam classifiers today work reasonably well, because there are lots of examples of both spam and nonspam emails.

Location: 1,236

So when we say life outcomes are hard to predict, we are using a combination of these three criteria: real-world utility, moral legitimacy, and irreducible error, that is, error that won't go away with more data and better computational methods.

Location: 1,248

Instead of predictions, the dominant approach in social science is to focus on improving our understanding of what causes phenomena of interest.

Location: 1,315

so far, we do not have theories about the predictability of social problems. We can't predict the future well, and we don't know what the fundamental limitations of our predictions are.

Location: 1,340

What if we can't make accurate predictions because there aren't enough people on Earth to learn all the patterns that exist?

Location: 1,427

All successful people are lottery winners to some extent. But they rarely acknowledge the role of luck in their success.

Location: 1,430

As E. B. White said, "Luck is not something you can mention in the presence of self-made men."

Location: 1,460

the market for cultural products has rich-get-richer dynamics built into it, also called “cumulative advantage.” Regardless of what we may tell ourselves, most of us are strongly influenced by what others around us are reading or watching, so success breeds success.

Location: 1,462

The depressing part is this: among the vast universe of “good enough” cultural products, it is a largely random process that determines success.⁴⁴ This is a mathematical consequence of cumulative advantage. The effect of an initial review of a book or rainy weather on the opening weekend of a film can get amplified over time.

Location: 1,512

Much of social media is a giant meme lottery. At any given time, there’s a certain finite appetite for memes. Memes work because they’re shared culture. In other words, the more a meme is shared, the more valuable it becomes, and the more likely people are to share it further.

Location: 1,965

Prior to the use of deep learning, computer vision researchers hand-coded algorithms for converting pixels into conceptually meaningful elements of images. All of that proved unnecessary and even counterproductive. But, yet again, the absence of experts also makes it harder to assess the behavior of AI systems on dimensions not captured by the benchmarks.

Location: 2,005

It turns out that the hard part of image classification is getting a model to learn the visual structure of the world—its patterns, objects, and so on. This is what a deep learning model uses most of its layers for—all but the final layer. The final layer is a simple process of converting those concepts to labels—the words or categories we use to describe images. At the penultimate layer, the model outputs a sequence of numbers, called a vector, that corresponds to a high-level description of the image. As shown in figure 4.6, two images of dogs will result in sequences of numbers that are close to each other.

Location: 2,063

Artists and photographers have had their work and style copied without compensation.^{41,42} Sometimes the generated images even include remnants of artist signatures.⁴³ Creators see none of the profits. The fear is that these tools will be used to replace rather than augment human artists,⁴⁴ especially for routine commissions like cover images and company logos.

Location: 2,087

Unlike predictive AI, which is dangerous because it doesn't work, AI for image classification is dangerous precisely because it works so well. The same technology that can be used for image classification and reverse image search also allows for mass surveillance. Image classification and face recognition are technically very similar.

Location: 2,179

To generate a single token—part of a word—ChatGPT has to perform roughly a trillion arithmetic operations. If you asked it to generate a poem that ended up having about a thousand tokens (i.e., a few hundred words), it would have required about a quadrillion calculations—a million billion. To appreciate the magnitude of that number, if every single person in the world together performed arithmetic at the rate of one calculation per minute, eight hours a day, a quadrillion calculations would take about a year. All that to generate one single response.

Location: 2,257

Chatbots “understand” in the sense that they build internal representations of the world through their training process. Again, those representations might differ from ours, might be inaccurate, and might be impoverished because they don't interact with the world in the way that we do. Nonetheless, these representations are useful, and they allow chatbots to gain capabilities that would be simply impossible if they were merely giant statistical tables of patterns observed in the data.

Location: 2,276

Philosopher Harry Frankfurt defined bullshit as speech that is intended to persuade without regard for the truth.⁶⁵ In this sense, chatbots are bullshitters.

Location: 2,354

The ability to have a conversation without getting bombarded with inappropriate outputs is a major reason for the viability and success of chatbots. To enable this, humans have to label millions of examples of toxic text and images. Labeling or annotating such content can be brutal. Annotators have to deal with low pay, high workloads, and exposure to intense content day in and day out. Because of cheaper labor and fewer regulations, much of this work happens outside the United States and Europe. Sama, one of the firms contracted by OpenAI in Kenya, pays workers between USD 1.46 and 3.74 per hour.

Location: 2,386

To improve the conditions of AI annotation work, an essay by Adrienne Williams and coauthors makes three recommendations: unionization, transnational organization, and solidarity between highly paid tech workers and their lower-paid counterparts.⁸⁶

Location: 2,419

Since the dawn of time, “seeing is believing” has been a reliable guide for seeking truth. Now, relatively suddenly, that rug has been pulled out from under us. Adapting to this new world won’t be easy.

Location: 2,445

It’s hard to grasp the possible consequences of AGI. Most jobs could be automated. In particular, one task that humans conduct today is AI research. If AGI took over AI research from humans, it could carry out AI research on its own and thus improve itself—over and over. And it could potentially do this much faster than humans can improve AI. The result would be artificial “superintelligence”: AI that not only matches but vastly exceeds human abilities across the spectrum.

Location: 2,466

We shouldn’t let the bugbear of existential risk distract us from the more immediate harms of AI snake oil.

Location: 2,590

FIGURE 5.3. The ladder of generality up until the early 2010s.

Location: 2,610

FIGURE 5.4. The ladder of generality at the time of writing.

Location: 2,662

Weizenbaum wrote, “I had not realized ... that extremely short exposures to a relatively simple computer program could induce powerful delusional thinking in quite normal people.” This has been termed the “ELIZA effect.”

Location: 2,693

by the time we achieve AI that surpasses human knowledge, yet more bottlenecks might be discovered. We can’t be sure what those will be, but we can guess. For example, much of the most prized and valuable human knowledge comes from performing experiments on people, ranging from drug testing to tax policy. This may mean that self-improvement can’t happen in a vacuum but would require AI to interact with the social world, not just the physical one. And it’s not clear how much you can speed that up.

Location: 2,740

Let’s define power as the ability to modify the environment (which is what makes the paper clip maximizer dangerous). Once we do so, a radically different picture emerges (figure 5.6). FIGURE 5.6. When we reframe the issue from “intelligence” to power, things look very different. Humans are powerful not primarily because of our brains but because of our technology.

Location: 2,782

But over time, the cost of training models at a given capability level decreases quickly. Training an image recognition classifier with the same performance became forty-four times cheaper between 2012 and 2019.¹⁷ And as we write this book, one of the most capable open-source language models can be trained for a cost of less than USD 1 million.¹⁸ To have any hope of stopping the development of powerful AI, governments will need to conduct draconian surveillance and achieve an unprecedented level of international cooperation.

Location: 2,876

on social media, content moderation is arguably the primary product. That's because the technical features behind platforms are easy to re-create. For example, an alternative social networking app called Mastodon offers much of the functionality of X (formerly Twitter)—and for a long time, the company had fewer than ten employees.

Location: 3,067

One consequence of cultural incompetence is that each platform's policies are homogeneous—the same everywhere in the world. And because most prominent social media platforms are U.S.-based, these policies tend to be too. U.S.-centric policies can be too strict in some ways and too permissive in other ways from other countries' perspectives. For example, banning nipples based on American sensitivities is seen as prudish by Europeans.³¹ On the other hand, Holocaust denial, which is illegal in most of Europe, was not prohibited by most platforms until recently.

Location: 3,090

Our guess is that if social media platforms took their international commitments seriously—if they tried to pay attention to local context all around the world to the same extent that they do in the United States and Europe—they would go out of business. In other words, they are able to offer a relatively polished product with a welcoming and reasonably well-enforced set of rules in Western countries only because they have an exploitative relationship with most non-Western ones.

Location: 3,142

Knowledge that overturns consensus is not an anomaly; it is how intellectual progress happens. So regardless of the technology behind misinformation detection, there is a risk of overreaching, resulting in a system that reinforces the political or scientific establishment while suppressing dissent. Fortunately, platforms don't try to remove all misinformation, but only certain categories of harmful misinformation.

Location: 3,434

Governments influence platform policies through the threat of regulation or being banned. The more important a country's market is to a company, the more the company cares about keeping the government happy. And the more authoritarian a country's government is, the more likely it is to demand censorship. These two criteria help explain why India and Turkey in particular have seen prominent clashes between governments and Western social media platforms.

Location: 3,462

The fact that the world changes over time is an intrinsic barrier to the effectiveness of the AI techniques used in content moderation: machine learning and fingerprint matching.

Location: 3,522

Researchers have published algorithms such as bridging-based rankings where content that brings people together is rewarded over content that pushes them apart.⁷¹ So far, no major platform has adopted such ideas. The only one to have tried is X (formerly Twitter). In 2022, the company launched Community Notes, a feature that allows users to correct misleading or incorrect information in posts.^{72,73} In an attempt to bridge the gap between people with differing political beliefs, notes that are accepted by people across the political spectrum are more likely to show up on posts. But the overall impact of Community Notes on the platform is small, since only a small fraction of posts are egregious enough to receive a note. And

critically, the feature doesn't change the algorithm that determines which posts show up on feeds, which continues to amplify divisive content.

Location: 3,560

"It is often more effective to misinform people implicitly by taking a true event (e.g., an adverse reaction to a vaccine, a rare instance of voter fraud, or a crime committed by a minority group) and suggesting it is representative of a broader pattern, than it is to make a verifiably false claim."⁷⁶ Some content is not harmful by itself but because of the effect it generates in some (but not other) people who see it. Social media has had a devastating effect on teens' mental health by encouraging social comparison, but good luck banning Instagram selfies. And as for hate speech, it is often more effective to incite violence against a group by spreading fear about that group than by calling for violence directly.

Location: 3,566

Another limitation of content moderation, which operates on a post-by-post basis, is that some content becomes harmful only in the aggregate. There have been reports of prominent tourist and religious sites being ruined by a horde of influencers, each looking to score views with their Instagram or TikTok videos.⁷⁸ The impact of any one tourist is minimal, but the competition among them results in a tragedy of the commons. Another example: a single individual calling out another for a (real or perceived) social transgression is an everyday occurrence, but when thousands of people dunk on one target, as often happens on social media, it can ruin lives and careers.

Location: 3,590

Daphne Keller, former Google general counsel and now a scholar at Stanford, argues that by making moderation so expensive, the law will force platforms to moderate less.⁷⁹ In our view, policymakers should recognize that flawed individual decisions made using AI aren't the problem so much as a symptom of deeper, underlying structural issues.

Location: 3,600

Reforming trillion dollar companies won't be easy, and the public square is too important to be left in this broken state. Are there alternatives outside of the Big Tech companies? One model worth looking at is Reddit. There, rather than one giant network of users, discussion is broken up into topic-focused "subreddits." The critical difference is that moderation is done by volunteer members of those subreddits rather than company employees or contractors.⁸¹ They are assisted by automated tools developed centrally by the platform. This model has many admirable qualities. Since moderators are members of the specific community that a post appears in, they are keenly aware of context rather than oblivious to it. It gives users a voice. If they don't like the moderation policies or enforcement, they can talk to the moderators or perhaps consider becoming a moderator. Furthermore, the Reddit model has the separation of functions that Douek recommends, with moderation being relatively independent from the company's commercial side.

Location: 3,615

The amount of moderation labor on Reddit appears to be several orders of magnitude lower than on Big Tech platforms.

Location: 3,746

The inadequacy of the Gartner hype cycle is even more sharply apparent when it comes to AI, because AI is an overarching term for many different technologies. Some types of AI have fundamental limits, such as those that claim to predict people's future. So even if the Gartner hype cycle were useful, different types of AI would be at different stages of it. In fact, AI research has historically seesawed between peaks and troughs instead of following the Gartner model.

Location: 3,770

it has gradually become clear that crypto is a solution looking for a problem.

Location: 3,817

In 2017, AI researchers Ali Rahimi and Benjamin Recht won the Test of Time Award at NeurIPS, one of the world's largest AI conferences. A Test of Time Award is awarded to papers that are ten years old and lead to significant impact on the field. After the award was announced, Rahimi was invited to give a speech. His twenty-minute speech was a scathing indictment of AI research,³² which he compared to alchemy due to the lack of rigor and low standards of evidence. Rahimi criticized the field's focus on beating previous results on benchmark datasets. Instead, he asked researchers to understand why AI tools perform well. "If you're building photo-sharing systems, alchemy is okay. But we're beyond that now. We're building systems that govern healthcare and mediate our civic dialogue," he said. "I would like to live in a society whose systems are built on top of verifiable rigorous thorough knowledge and not on alchemy."

Location: 3,900

When scientific fields have tried to test reproducibility systematically, they have found that many peer-reviewed studies fail to reproduce. Perhaps the most prominent example comes from psychology. In 2015, a large group of researchers tried to replicate past research in social psychology. They found that only 36 percent of published results could be replicated, despite being peer reviewed and published in top scientific journals. When other fields have tried to replicate past findings, they have found similar results: many studies fail to reproduce.

Location: 3,906

In a 2018 study, Odd Erik Gundersen and Sigbjørn Kjensmo from the Norwegian University of Science and Technology set out to investigate the reproducibility of AI research. They reviewed four hundred papers from leading AI publications to ascertain if they contain enough detail to be reproducible by an independent researcher. They found that none of the four hundred papers satisfied all of the criteria (such as sharing their code and data) for reproducibility. Most papers satisfied merely twenty to thirty percent of the reproducibility requirements they identified, making it hard to even investigate if the results were reproducible.

Location: 3,922

a cardinal rule of AI: never test on the training data.

Location: 3,923

Leakage is well known in AI. In an apocryphal story from the early days of computer vision, a classifier was trained to discriminate between images of Russian and American tanks with seemingly high accuracy. It turned out, however, that the Russian tanks had been photographed on a cloudy day and the American ones on a sunny day; the classifier was not detecting the difference between the tanks at all, but rather had merely learned to detect the brightness of the image.

Location: 3,927

Since AI is increasingly used for scientific research, how often does leakage affect results in other disciplines? We reviewed the academic literature to find results similar to ours. It turned out that there was no shortage of errors due to leakage in AI-based science. Hundreds of papers in over a dozen scientific disciplines—including medicine, psychiatry, computer security, IT, and genomics—had all been affected by leakage.

Location: 3,940

Another reason for lack of reproducibility is researchers' reliance on commercial AI models.

Location: 3,997

When describing results from academic research on AI, news reports often include accuracy numbers. For instance, a 2022 Bloomberg article on a crime prediction study was titled "Algorithm Claims to Predict Crime in US Cities before It Happens."⁵⁹ The article said the study touted 90 percent accuracy for the model. We've already seen how developers make their accuracy numbers look good by simply changing the metric they are evaluated on (top-3 versus top-5 accuracy). Similarly, researchers have many ways to make their predictions look

good. In this study, authors had a one-day margin of error in their predictions (they marked a prediction as correct if a crime occurred on the day they predicted, or one day before or after the prediction). But there is rarely enough space in a news article to explain the context for how estimates of model performance like accuracy are calculated or what they represent.

Location: 4,058

Describing AI as unknowable reduces our agency by positioning AI as something that we can never understand and therefore never challenge. Besides, the most important questions about AI are not about its internals. For example, to investigate the accuracy of Epic's sepsis prediction tool, or any of the other examples of snake oil we've seen, knowing how the system works internally isn't necessary—what is needed is information about how predictions by the model turned out.

Location: 4,080

testing chatbots on benchmarks designed for humans tells us little about whether they can automate jobs in the real world.

Location: 4,084

The real impact of AI is likely to be subtler: AI will shift power away from workers and centralize it in the hands of a few companies.

Location: 4,178

If we keep going down the path of AI as almost entirely private and profit driven rather than guided by public interest, the risks are clear.

Location: 4,180

We must first recognize that much of the downside of AI comes down to factors outside the technology itself—like the incentives of the institutions that use AI.

Location: 4,189

The demand for AI snake oil here isn't primarily about AI—it's about misguided incentives in the failing institutions that adopt them.

Location: 4,190

For example, if the state of hiring weren't so broken, and we had something resembling a decent way to match candidates to jobs, would hiring managers still rely on HireVue? For hiring managers who need to go through hundreds or perhaps thousands of candidates to fill a single position, using HireVue can be enticing, even though it filters candidates based on questions like "Do you keep your desk neat or untidy?"

Location: 4,268

We can aim to find strategies or policies that achieve modest efficiency gains while being simple enough to understand—both for decision-makers and decision subjects. Simplicity helps decision-makers assure themselves that things can't go catastrophically wrong and build trust with decision subjects. Such an approach also makes it easier to incorporate multiple objectives, some of which capture moral rather than economic goals. One example in the criminal justice system is showing leniency toward younger defendants, on the basis that they are less morally culpable for their actions, even if they are statistically more likely to reoffend.

Location: 4,273

To illustrate what an alternative decision-making approach might look like, consider partial lotteries. Instead of trying to pick the "top" applicants who would, say, receive a grant or get into a college, partial lotteries make randomness an explicit part of the decision-making process. All applicants who satisfy a certain basic cutoff are included in a pool, and a random draw is used to pick the ones who get in. Our point is not that partial lotteries are always the right answer, but that radically different strategies might be worth considering in some cases.

Location: 4,303

Especially in the face of resource scarcity, partial lotteries offer a decision-making mechanism that reduces rich-get-richer feedback loops, helps us acknowledge the role of randomness in decision-making systems, reduces wasted time and effort in applications, and allows us to study the effects of decisions.

Location: 4,327

In the United States, AI regulation is vertical. That is, instead of having a broad agency that is tasked with governing all AI products and services, federal agencies have the authority within specific sectors for regulating AI. There are hundreds of such agencies. For example, the Consumer Finance Protection Bureau protects consumers in the financial marketplace from deceptive claims and discrimination, whereas the Food and Drug Administration regulates AI used in medical settings.

Location: 4,335

In contrast, the EU has come up with horizontal rules for regulating AI—rules that apply across many sectors. There are different laws that regulate different aspects of AI: The General Data Protection Regulation (GDPR) regulates how companies collect, store, and use personal data. It is relevant to AI systems because it aims to ensure that AI respects privacy. The Digital Services Act (DSA) requires transparency and audits on the use of AI on online platforms and social media. The Digital Markets Act (DMA) is aimed at increasing competition in online platforms, such as by disallowing large online platforms from self-preferencing their own results. Most notably, the Artificial Intelligence Act (AIA) includes a risk-based taxonomy of AI applications. Developers building applications that are high risk (such as AI used for hiring, educational access, or worker management) need to comply with many transparency requirements.

Location: 4,343

China has a third approach that has elements of both of the above.³¹ The country began its oversight of AI with vertical rules, including requirements for transparency around online recommender systems in 2017 and "deep synthesis" systems, such as AI used to create images, video, and text, in 2022. After the release of ChatGPT and the overwhelming public

interest in generative AI, regulators released a new set of draft regulations on generative AI in April 2023, which focused on text-based systems. A key part of their regulations is that content produced using AI should embody “Core Socialist Values.” Chatbots are no doubt an important avenue through which the Chinese government will try to control what information people can access and what they are allowed to say.

Location: 4,361

The notion that self-regulation is the only realistic option because of the slow pace of regulation is also based on the flawed premise that only tech companies can understand and manage technology. But as we’ve seen throughout this book, the principles of AI systems being used today are simple enough to be broadly understood.

Location: 4,366

The fact is that policymakers don’t need domain expertise. They delegate all the details to experts who work at various levels of government and in various branches. The two of us have been fortunate enough to consult with many of these experts, and they tend to be extremely competent and dedicated. Unfortunately, there are too few of them, and the understaffing of tech experts in government is a real problem.

Location: 4,409

Regulatory capture happens when regulators are either misinformed or lack the resources and funding to function independently of the companies they are regulating. So, the best way to avoid regulatory capture is to strengthen existing regulators and provide them with the funding and resources to operate independently.